

Soohwan Lee

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Human-centered AI | Human-AI Collaboration | Conversational AI | Group Dynamics

EDUCATION

- **Ulsan National Institute of Science and Technology(UNIST)** Present
Ph.D in Design with Concentration in HCI Ulsan, South Korea
 - Advisor: Kyungho Lee
- **Ulsan National Institute of Science and Technology(UNIST)** 2022
M.S in Creative Design Engineering with Concentration in HCI Ulsan, South Korea
 - Committee: Kyungho Lee(Chair), Ian Oakley, Young-woo Park
- **Ulsan National Institute of Science and Technology(UNIST)** 2020
B.S in Human Factors Engineering & Computer Science and Engineering Ulsan, South Korea

EXPERIENCE

- **Ulsan National Institute of Science and Technology (UNIST)** Ulsan, South Korea
 - **Research Assistant** 03/2020 – Present
Expressive Computing Lab, UNIST (Advisor: Kyungho Lee)
 - **Research Intern** 12/2018 – 12/2019
Interactions Lab, UNIST (Advisor: Ian Oakley)

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=SHORT PAPER AND POSTER, T=THESIS, U=UNDER REVIEW

[Peer-reviewed Conference Proceedings]

- [C.1] **Soohwan Lee**, Seoyeong Hwang, Ian Oakley, and Kyungho Lee. (2024). **Expanding the Design Space of Vision-based Interactive Systems for Group Dance Practice**. In Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24)
- [C.2] DoYoung Lee, **Soohwan Lee**, and Ian Oakley. (2020). **Nailz: Sensing Hand Input with Touch Sensitive Nails**. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)

[Journal Publications]

- [J.1] Jang, Sangsu, **Soohwan Lee**, Temirlan Dzhoroiev, Taeyoon Kim, Hyejin Oh, Nari Kim, and Young-Woo Park (2021). **Design Guidelines for Contextual Awareness and Management of Hygiene in Daily Life with Infectious Viruses**. Archives of Design Research

[Short Papers, Posters, and Workshop Papers]

- [S.1] **Soohwan Lee**, Seoyeong Hwang, and Kyungho Lee. (2025). **Beyond Individual UX: Defining Group Experience(GX) as a New Paradigm for Group-centered AI**. In Companion Publication of the 2025 ACM Designing Interactive Systems Conference (DIS'25 Companion)
- [S.2] **Soohwan Lee**, Seoyeong Hwang, Dajung Kim and Kyungho Lee. (2025). **Conversational Agents as Catalysts for Critical Thinking: Challenging Social Influence in Group Decision-making**. In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)
- [S.3] **Soohwan Lee**, Mingyu Kim, Seoyeong Hwang, Dajung Kim and Kyungho Lee. (2025). **Amplifying Minority Voices: AI-Mediated Devil's Advocate System for Inclusive Group Decision-Making**. In Companion Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25 Companion)
- [S.4] **Soohwan Lee**, Seoyeong Hwang, and Kyungho Lee. (2024). **Conversational Agents as Catalysts for Critical Thinking: Challenging Design Fixation in Group Design**. (DIS '24 Workshop)
- [S.5] **Soohwan Lee**, and Kyungho Lee. (2023). **CheerUp: A Real-time Ambient Visualization of Cheerleading Pose Similarity**. In Companion Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23 Companion)

[Thesis]

[T.1] Soohwan Lee. (2022). **NEAS: AI-based Interactive System to Support Non-verbal Expressions in Online Communication**. Master Thesis, UNIST.

[Under Review (TBA)]

[U.1] Soohwan Lee, Seoyeong Hwang, Mingyu Kim, Dajung Kim, and Kyungho Lee. TBA. **Investigating LLM-Powered Minority Support in Power-Imbalanced Group Decision-Making: Counterargument and Mediation as Intervention Strategies**. (Conference Paper)

[U.2] Soohwan Lee and Kyungho Lee. TBA. **Understanding Compliance and Conversion Dynamics in Multi-Agent Collectives**. (Conference Paper)

[U.3] Seoyeong Hwang, Soohyun Hwang, Soohwan Lee, Dajung Kim, and Kyungho Lee. TBA. **Augmentary: Exploring How LLM-Based Journaling Systems Can Support Reflective Writing and Autonomous Meaning-Making**. (Conference Paper)

[U.4] Soohwan Lee, Soobin Lee and Kyungho Lee. TBA. **Expression2Emoji: Designing Non-verbal Communication Support Tool for Group Video Meetings**. (Conference Paper)

PROJECTS

• Commercialization design and development of Intelligent Product-Service System for personalized full silver life cycle care <i>This work was supported by the Technology Innovation Program (20015056, Commercialization design and development of Intelligent Product-Service System for personalized full silver life cycle care) funded By the Ministry of Trade, Industry & Energy(MOTIE, Korea).</i>	01/2022 - Present
• Develop sensor modules and mountings for monitoring upper body angles and angular velocities <i>This work was supported by LG Electronics.</i>	01/2024 - 09/2024
• Building a end-to-end speech recognition model for active seniors using KoSpeech model <i>This research was supported by X-Corps Plus program of National Research Foundation of Korea(NRF) funded by the Ministry of Science, ICT (No. 2021H1D8A306520712).</i>	03/2023 - 12/2023
• Smart Clean BioAir Home Management System <i>This work was supported by the Technology Innovation Program (20009977, Smart Clean BioAir Home Management System) funded By the Ministry of Trade, Industry & Energy(MOTIE, Korea).</i>	04/2020 - 12/2021
• UNIVERSE: Design of a Social Robot that Supports Emotional Well-Being of the MZ Gen <i>This work was supported by Innovative Education Center, UNIST. (This project is part of AI Challenger Program at UNIST.)</i>	09/2021 - 12/2021
• Conceptualize and design a new UX/UI for a washer/dryer targeting Gen MZ. <i>This work was supported by LG Electronics.</i>	03/2021 - 09/2021

TEACHING

• Teaching Assistant <i>UNI207::Creative Computing for Emerging Media ART, UNIST</i>	07/2022 - 08/2022
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HONORS AND AWARDS

• CES Innovation Awards <i>2025 Honoree in Smart Home</i>	2025 
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INVITED TALKS

• Collaborating with AI in Groups: A Design Research Approach to Social Influence <i>Group Dynamics & Social Change Lab, Lehigh University (Hosted by Dr.Jiin Jung)</i>	04/2025
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ACADEMIC SERVICES

Have reviewed 13 submissions in total, including full papers and posters.

3 Special Recognitions for Outstanding Reviews: UIST'25 Papers, DIS'25 Papers, CHI'25 LBW

• ACM DIS PWiP Associate Chair	2025
• ACM DIS Paper Reviewer	2025
• ACM CSCW Paper Reviewer	2024
• ACM UIST Paper Reviewer	2025
• ACM CHI LBW Reviewer	2025
• ACM CSCW Poster Reviewer	2025
• ACM CUI WiP Reviewer	2025
• Handbook of Computational Social Psychology Reviewer	2025